

JOSE ELMAR ACUNO

Software Engineer | Embedded Systems & IoT | Machine Learning | Process Automation

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PROFESSIONAL SUMMARY

Computer Engineering graduate with a builder's mindset spanning embedded systems, IoT, machine learning, and business process automation. Currently automate manual, high-volume data reconciliation for a national real estate developer, replacing a five-day manual process with a one-click Python tool of my own design. Background includes ESP32-based IoT development, SVM-based crash detection research, RFID access-control systems, and enterprise IT support. Comfortable moving across hardware, software, and data and drawn to environments where I can keep learning and applying new technology, including AI, as a force multiplier rather than a shortcut.

EXPERIENCE

Commission Processing Assistant - SM Development Corporation

Mar 2026 – Present

- Consolidate and reconcile high-volume, multi-source commission and sales entries in Excel, resolving duplicates and ensuring figures tally against expected sales.
- Identified inefficiencies in the manual reconciliation workflow and independently designed and built Python automation tools “Auto Consolidator” and “Auto Double Checker” that reduced a 5-day manual cleanup process to a single click.
- Applied an engineering approach to a non-technical process, turning a repetitive manual task into a repeatable, self-maintained automation pipeline.
- Continuously refine and extend automation scripts as data structures and business rules evolve.

Freelance Software & Hardware Consultant - Self-Employed

Jul 2025 – Present

- Deliver custom software, embedded hardware, and small-scale system builds for private clients and student project requests, spanning both hardware prototyping and full-stack development.
- Advise clients on system design, feasibility, and implementation for hardware and software projects.
- Use AI tools deliberately to accelerate tool-building and prototyping, rather than as a substitute for engineering judgment.

EUC (End User Computing) Intern - Straive (formerly SPi Global)

Mar 2025 – May 2025

- Provided Level 1 IT support through calls and tickets: managed IT asset inventory, receiving and tagging laptops and equipment.
- Prepared systems for deployment Windows installation, imaging, drivers, and Linux and performed RAM/SSD/HDD troubleshooting and reformatting.
- Applied BitLocker encryption and installed enterprise security tools; conducted remote troubleshooting using Kaseya and PCVisor.
- Assisted with Active Directory administration and basic network-related tasks.

Electrical Engineer Intern - First Laguna Electric Cooperative Inc. (FLECO)

Dec 2019 – Feb 2020

- Climbed electrical poles and operated aerial lift devices to access and repair electrical infrastructure.
- Assisted in inspection, maintenance, and basic troubleshooting of electrical systems.
- Reviewed design drawings and technical specifications for newly engineered components, ensuring compliance with customer requirements and safety standards.

TECHNICAL PROJECTS

Rider Rescue - Motorcycle Crash Detection (IoT + Machine Learning)

- Developed an IoT-based safety device using ESP32 and LILYGO T-A7670E (GSM/GPS) with real-time location tracking uploaded to Firebase via GSM/GPRS.
- Integrated MPU9250, ADXL345, and GPS sensors to build crash-detection logic based on acceleration, orientation, and speed changes.
- Applied a Support Vector Machine (SVM) model to classify crash vs. non-crash events and trigger automatic SMS alerts.
- Deployed the device for real-world field testing and data collection; results published as a research paper (see below).

RFID Monitoring System

- Designed a full-stack RFID-based monitoring and access-control system using HTML, CSS, JavaScript, and Firebase Realtime Database.
- Implemented user authentication, admin approval workflow, role-based access, and real-time dashboards with attendance tracking and duration analytics.
- Integrated EmailJS notifications and live Firebase listeners; deployed via Firebase Hosting.

TECHNICAL SKILLS

- Programming & Automation: Python (process automation & scripting), C/C++, JavaScript, PHP
- Embedded & IoT: ESP32, Arduino, Raspberry Pi, GSM/GPRS, GPS/GNSS, RFID
- Machine Learning: Support Vector Machines (SVM), MATLAB, classification modeling
- Data & Productivity: Advanced Excel (reconciliation & automation), Firebase Realtime Database
- Web: HTML, CSS, JavaScript, Google Apps Script
- IT & Systems: Windows Deployment, BitLocker, Active Directory, Linux
- Tools: VS Code, Arduino IDE, Firebase Console, Git

RESEARCH PUBLICATION

"RiderRescue: Motorcycle Crash Detection for Emergency Response Using Support Vector Machine Algorithm in MATLAB"

- Presented at the IEEE International Conference on Advanced Research in Computing, 2025.

ROBOTICS & COMPETITIONS

Mechatronics and Robotics Society of the Philippines - National Robotics Competition

- Competed with automated and line-following robots, applying sensor logic, motor control, and algorithmic problem-solving.
- Participated in breadboarding ICs and logic circuit implementation.

CERTIFICATIONS

- National Certificate II - Computer System Servicing, 2025
- Cognizant Generative AI, 2025
- MRSP National Robotics Competition Certification, 2025
- MRSP IoT Training Certification, 2025

EDUCATION

Bachelor of Science in Computer Engineering

National University Laguna - 2025